Joseph Mesches

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Electrical Engineering student (B.S., expected 2027) and board-certified critical-care flight paramedic. Engineering focus on biomedical instrumentation, signal processing, embedded systems, and machine learning; prior career delivering critical care across aeromedical, emergency-department, and prehospital settings.

Education

Bachelor of Science, Electrical Engineering — Aug 2023 – Expected May 2027
Colorado State University — *Fort Collins, CO*

- Major GPA 3.7 | Cumulative GPA 3.4 | Dean's List | IEEE-Eta Kappa Nu (HKN) | Walker-Weinshenker Family Scholar.
- Coursework: Biomedical Signal & Image Processing, Linear Systems, Communications, Electromagnetic Fields, Microcontrollers & IoT, Digital Logic, Digital System Design, Electronics, Circuit Theory, Differential Equations.
- In progress (Fall 2026): ECE 512 Digital Signal Processing, ECE 528 Embedded Systems / ML, ECE 541 Applied Electromagnetics.

Electrical Engineering Technology (transfer coursework) — Jan 2023 – July 2023
Metropolitan State University of Denver — *Denver, CO*

Associate of Applied Science, Paramedicine — 2016 – 2019
Arapahoe Community College — *Littleton, CO*

Technical Skills

Programming: C/C++, embedded C, Python (NumPy, SciPy, Pandas, Matplotlib, scikit-learn, PyTorch, Keras), MATLAB, Verilog, ARM Assembly, Java, SQL, Bash, LaTeX
Machine Learning & DSP: classification, ensemble learning, explainable AI / feature attribution, 1D CNNs, adaptive filtering, FFT & windowing, wavelet transforms, QRS / fiducial detection, feature extraction, real-time embedded DSP, edge inference (Coral TPU), vision-language models, RAG
EDA & Simulation: Altium Designer, KiCad, LTSpice, Multisim, Cadence Virtuoso, Quartus Prime
Biomedical Instrumentation: 12-lead ECG/EKG, EEG, analog biopotential front-ends, patient monitor/defibrillators, ventilators, infusion pumps, intra-aortic balloon pumps
RF & SDR: RTL-SDR, I/Q sampling, ADS-B, GNU Radio, spectrum analyzers, VNAs, antenna deployment
Lab & Protocols: oscilloscopes, logic analyzers, function generators, FPGAs, soldering, microsoldering; SPI, I2C, UART, CAN, JTAG, MQTT
Software & Infrastructure: Git, Linux systems administration (5+ years), Docker, Proxmox, Grafana, Jupyter, relational databases, web interfaces

Engineering & Technical Experience

Student Research Assistant, Radar Systems — Aug 2026 – Present
Prof. Chandrasekar's Radar Lab, Colorado State University — *Fort Collins, CO*

- Developing AQPI Sentinel, a monitoring interface for radar products, and integrating it with the lab's backend systems.

Electrical Engineering Intern — June 2026 – Present
The RMH Group, Inc. (MEP) — *Lakewood, CO*

- Reverse-engineered an automatic racking robot and conveyor system into AutoCAD drawings for a toner-manufacturing facility.
- Contributed to multiple control-system projects to improve operational efficiency.

– Maintained and updated as-built drawing packages (AutoCAD 2024, Adobe Acrobat Pro) to ensure accuracy for maintenance use; invited to continue with the firm through the final academic year.

ECE CreatorSpace Lab Assistant Nov 2023 – May 2024
Colorado State University *Fort Collins, CO*

– Supported students with PCB design, embedded systems, and lab equipment across the ECE makerspace.

CUSP Research Intern Summer 2025
Webb-Waring Institute, CU Anschutz *Aurora, CO*

– Developed and executed an assay for oxidative-burst formation in immune cells; performed data analysis and interpretation of results.

Selected Projects

OpenOrbit Radar Payload Team (Senior Design) Aug 2026 – May 2027
Project Manager, Colorado State University

– Developing a drone-based synthetic aperture radar (SAR) imaging system as part of the capstone engineering team.
– As project manager, manage schedule and deliverables, budget, funding and sponsor relations, and team resourcing.

Explainable ECG Clinical Decision-Support System Spring 2026
ECE 431, Colorado State University

– Clinical decision-support system classifying 12-lead ECGs into five diagnostic superclasses on the PTB-XL dataset; a five-pipeline ensemble (XGBoost meta-classifier over engineered signal-processing features plus a 1D-ResNet CNN) reaching 0.93 macro-AUC, on par with published deep-learning benchmarks and validated on a second hospital dataset without retraining.
– Integration lead on a 4-person team: authored the shared ~5,500-line DSP library (filtering, Pan-Tompkins R-peak detection, QRS/P/T delineation), the morphological feature pipeline, and the ensemble harness. Predictions trace to named clinical criteria (Sokolow-Lyon, Cornell, Sgarbossa, Romhilt-Estes) for auditable, clinician-reviewable output.

Rodent Activity Monitor Fall 2025
CU Anschutz Neurotechnology Center

– Custom ESP32 PCB (PIR, CO2, temperature, humidity, battery management); 5 deployed units, 96-hour to 14-day battery life on configurable sampling.
– Full-stack pipeline (Docker, database, web UI, Grafana) in daily use by lab non-engineers.

Edge ML Wildlife Tracker 2024 – Present
Personal

– Low-latency on-device detection using IP cameras and a Coral Edge TPU; integrated vision-language models for behavioral classification and pattern-of-life logging with mobile alerts.

Self-Hosted LLM Infrastructure 2024 – Present
Personal

– Deployed a self-hosted LLM stack with a vector RAG database over course materials and a shared web interface; engineered system prompts for reliable engineering-task outputs.

Low-Frequency Spectrum Analyzer Spring 2024
Colorado State University

– Led a 3-person team building a battery-powered ESP32 handheld with TFT display; implemented FFT and windowing in C++ for 1–2 Hz resolution.

Evoked Potential TTL Controller 2022
CU Anschutz Neurotechnology Center

– Sole engineer: custom Arduino-based PCB and enclosure to trigger visual evoked-potential hardware.

IV Fluid Warmer Spring 2022
Personal

– Arduino-based resistive fluid warmer with adaptive PID temperature control.

Multi-Channel EEG Analysis Tool 2021
CU Anschutz Neurotechnology Center

– Python pipeline (NumPy, Pandas, Matplotlib) for multi-channel EEG; adaptive filtering and frequency analysis for artifact removal and feature extraction.

ADS-B Passive RF Collection Node 2023 – Present
Personal

– 24/7 RTL-SDR passive collection node at 1090 MHz feeding community aggregation networks; antenna siting, grounding, and uptime management.

Self-Managed Homelab / Hypervisor Cluster Ongoing
Personal

– Six-node Proxmox cluster running containerized services (Docker, Grafana, databases) on Linux.

Clinical & EMS Experience

Flight / Critical Care Paramedic Mar 2022 – Nov 2022
UCHealth LifeLine *Colorado*

– Delivered critical care as Pilatus PC-12 flight crew to the most acutely ill and injured, including adult and pediatric trauma, medical-ICU, and high-risk maternal-fetal (obstetric) patients, across scene response, emergency department, ICU, and inter-facility transport.
– Managed advanced physiological monitoring: 12-lead ECG; invasive arterial, pulmonary, and intracardiac pressures; intracranial pressure transduction; waveform and quantitative capnography; pulse oximetry, plethysmography, and co-oximetry; and cardiotocography. Initiated and troubleshot advanced mechanical ventilation.
– Performed advanced procedures: direct and video intubation, surgical cricothyrotomy, chest tubes, central, intraosseous, and umbilical-vein access, and obstetric procedures; applied emergency and critical-care pharmacology and interpreted point-of-care labs (arterial and venous blood gases, BMP/CMP, CBC with differential, glucometry, and cardiac markers).

Flight / Critical Care Paramedic June 2021 – Dec 2021
Air Methods Corporation *Englewood, CO*

– Flight critical-care paramedic and base Field Safety Representative; managed intra-aortic balloon pumps, mechanical ventilators, and infusion pumps.

Assistant Instructor, EMS Academy May 2021 – Feb 2022
Arapahoe Community College *Littleton, CO*

– Delivered skills instruction under a primary instructor; coached technical communication and clinical practice.

Emergency Department Paramedic Feb 2020 – June 2021
UCHealth *Aurora, CO*

– Emergency-department paramedic at a Level-1 Trauma Center.

Field Training Officer / Critical Care Paramedic Jan 2018 – June 2020
Northglenn Ambulance *Commerce City, CO*

– 911 and interfacility critical care; field-trained new paramedics; QA and education committee member; authored training manuals and SOPs.

Paramedic / Field Training Supervisor Aug 2017 – Feb 2018
Mile High Ambulance, LLC *Greater Denver Area*

– ALS interfacility care; co-developed the agency field-training program.

Training Supervisor / EMT-IV Nov 2015 – May 2017
Capital City / Citywide Ambulance, LLC *Aurora, CO*

– Senior field-training officer for a BLS interfacility agency; built and ran the field-training program.

Sergeant May 2012 – Aug 2016
Arapahoe Rescue Patrol, Inc. *Littleton, CO*

– Leadership role in a search-and-rescue agency; planned and led training and missions; drafted training and regulatory policy.

Leadership & Service

IEEE Student Branch May 2025 – Present
Colorado State University *Fort Collins, CO*

– President (2025–2026); Vice President (2026–Present). Lead branch operations, outreach, and communication between students, faculty, and industry partners.

Publications

Mesches, J. "Looking Past Dementia Reveals Hidden Life Threats." *Journal of Emergency Medical Services* (JEMS), 2018.

Certifications & Licenses

Active

– Certified Flight Paramedic (FP-C), ICAPP (2018; exp. 2030; ID 0008013)
– Nationally Registered Paramedic, NREMT (exp. 2028)
– Certified Paramedic, Critical Care Endorsement (P-CC), Colorado DPHE (exp. 2026)
– Colorado EMS Primary Instructor, Arapahoe Community College (2019)
– FEMA Independent Study: IS-5, IS-100, IS-200, IS-405, IS-700, IS-800, FEMA (2012)

Prior

– Certified Paramedic, Arizona DHS and California EMS Authority (2021–2023)
– Advanced Cardiac Life Support (ACLS) and Pediatric Advanced Life Support (PALS), American Heart Association
– Basic Life Support for Healthcare Providers and BLS Instructor, American Heart Association
– Prehospital Trauma Life Support (PHTLS) and PHTLS Instructor, NAEMT
– Transportation Provider Advanced Trauma Course (TPATC), ASTNA
– Neonatal Resuscitation Provider (NRP), American Academy of Pediatrics
– S.T.A.B.L.E. Provider, The STABLE Program
– Data or Specimens Only Research, CITI Program (2023–2026)
– Rescue Diver, PADI (2014)

Honors & Awards

– Walker-Weinshenker Family Scholar (2024–2026)
– Dean's List (2023)
– IEEE-Eta Kappa Nu (HKN) Member
– Compassionate Care Award (2018)
– Community Service Award